

Here is a master cheat sheet of all the essential terminal commands we used to build and customize your Raspberry Pi server.

You can save this list to your notes so you never have to memorize them!

1. File Editing Commands (The Nano Editor)

Use these whenever you want to change the Python code, update text templates, or change the API tokens.

- **Edit the Business/Print Bot:**

```
Bash
nano /home/redwannabil/businessbot.py
```

- **Edit the Admin Control Bot (Power & Performance):**

```
Bash
sudo nano /home/redwannabil/pi_control_bot.py
```

- **Edit the Master Background Service File:**

```
Bash
sudo nano /etc/systemd/system/telegram-notify.service
```

*(Remember: Inside Nano, you always save and exit by pressing **Ctrl + X**, then typing **Y**, and hitting **Enter**).*

2. Background Service Commands (Systemctl)

Use these to control the invisible background processes. You *must* run the restart commands every time you save changes to a Python file so Linux learns the new code!

- **Restart the Admin Control Bot:**

```
Bash
sudo systemctl restart telegram-notify
```

- **Restart the Business Print Bot:**

```
Bash
sudo systemctl restart printerbot
```

- **Check the Health/Status of a Bot (If it stops replying):**

```
Bash
sudo systemctl status telegram-notify
sudo systemctl status printerbot
```

- **Stop a Bot (To pause it or fix an infinite loop):**

```
Bash
sudo systemctl stop telegram-notify
sudo systemctl stop printerbot
```

- **Apply systemd Changes (Run this if you ever edit the .service file):**

```
Bash
sudo systemctl daemon-reload
```

3. Debugging Commands (The "Magnifying Glass")

If a bot stops replying and you need to see exactly what line of code is crashing, stop the background service first, then run one of these to force the error to print on your screen:

- **Test the Admin Control Bot manually:**

```
Bash
sudo /usr/bin/python3 /home/redwannabil/pi_control_bot.py
```

- **Test the Business Print Bot manually:**

```
Bash
sudo /usr/bin/python3 /home/redwannabil/businessbot.py
```

(Remember: Press **Ctrl + C** to stop the manual test and get your terminal prompt back).

4. Installation Commands (For future reference)

If you ever wipe your SD card and start over, these are the exact commands we used to give the Root administrator the tools it needed:

- **Install Telegram API for Root:**

```
Bash
sudo pip3 install pyTelegramBotAPI --break-system-packages
```

- **Install Performance Monitor Tools:**

```
Bash
sudo pip3 install psutil speedtest-cli --break-system-packages
```

Would you like me to compile the *complete, final Python code* for both of your bots into one message as well, just in case you ever need a clean backup?